

Numerical Solution of the One-Dimensional Heat Equation Using Explicit Finite Differences

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Abstract

In this paper, the one-dimensional heat equation is solved numerically using an explicit finite difference method. We derive the discretization scheme, discuss the von Neumann stability condition, and implement the method in Python. The numerical results are compared with the analytical solution, and the order of convergence of the method is empirically verified.

1 Introduction

The heat equation is one of the fundamental partial differential equations of mathematical physics. It describes the temporal evolution of the temperature distribution $u(x, t)$ in a medium and, in its one-dimensional form, is given by

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}, \quad x \in [0, L], \quad t > 0, \quad (1)$$

where $\alpha > 0$ denotes the thermal diffusivity of the material. Together with the boundary conditions

$$u(0, t) = u(L, t) = 0 \quad (2)$$

and the initial condition $u(x, 0) = u_0(x)$, the problem is well-posed. Although an analytical solution using Fourier series exists for many initial conditions, a numerical method is necessary for more complex geometries or variable coefficients $\alpha(x)$. In the following, we consider the simplest such method: the explicit finite difference method (FTCS, *forward-time centered-space*).

2 Discretization

We divide the spatial interval $[0, L]$ into N equal sub-intervals of length $\Delta x = L/N$ and the time axis into steps of length Δt . Let the grid points be $x_i = i\Delta x$ for $i = 0, \dots, N$ and $t_n = n\Delta t$. We approximate $u(x_i, t_n) \approx u_i^n$. The time derivative is replaced by a forward difference quotient,

$$\left. \frac{\partial u}{\partial t} \right|_i^n \approx \frac{u_i^{n+1} - u_i^n}{\Delta t}, \quad (3)$$

and the spatial second derivative by the central difference quotient

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_i^n \approx \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}. \quad (4)$$

Substituting into equation (1) yields the explicit update scheme

$$u_i^{n+1} = u_i^n + r(u_{i+1}^n - 2u_i^n + u_{i-1}^n), \quad r := \frac{\alpha \Delta t}{\Delta x^2}. \quad (5)$$

A von Neumann stability analysis shows that scheme (5) is stable if and only if

$$r = \frac{\alpha \Delta t}{\Delta x^2} \leq \frac{1}{2} \quad (6)$$

holds. This condition significantly restricts the time step Δt when using fine spatial resolution, as $\Delta t \propto \Delta x^2$ is required.

3 Implementation

The method from equation (5) can be implemented directly in Python. The following code shows a vectorized implementation using NumPy, which enforces the boundary conditions at each time step:

Listing 1: Explicit FTCS method for the heat equation

```
import numpy as np

def loese_waermeleitung(u0, alpha, dx, dt, n_schritte):
    """Solves the 1D heat equation explicitly."""
    r = alpha * dt / dx**2
    if r > 0.5:
        raise ValueError(f"Unstable: r = {r:.3f} > 0.5")

    u = u0.copy()
    for n in range(n_schritte):
        u_neu = u.copy()
        u_neu[1:-1] = u[1:-1] + r * (u[2:] - 2*u[1:-1] + u[:-2])
        u_neu[0] = 0.0 # Dirichlet boundary condition
        u_neu[-1] = 0.0
        u = u_neu
    return u
```

The stability condition is explicitly checked in line 6, ensuring that unstable parameter combinations are intercepted early.

4 Results

As a test case, we choose $L = 1$, $\alpha = 1$, and the initial condition $u_0(x) = \sin(\pi x)$, for which the analytical solution

$$u_{\text{exakt}}(x, t) = e^{-\pi^2 \alpha t} \sin(\pi x) \quad (7)$$

is known. Figure 1 shows the numerical solution at different time points compared to the analytical solution. To verify the order of convergence of the method, we calculate the maximum error $E_N = \max_i |u_i^{\text{end}} - u_{\text{exakt}}(x_i, t_{\text{end}})|$ for different spatial resolutions N with a fixed stability constant $r = 0.4$. Table 1 summarizes the results.

As expected, the method shows the theoretically predicted order of convergence $p \approx 2$ in Δx , since both the central difference quotient in equation (4) and the time integration at constant r provide quadratic convergence.

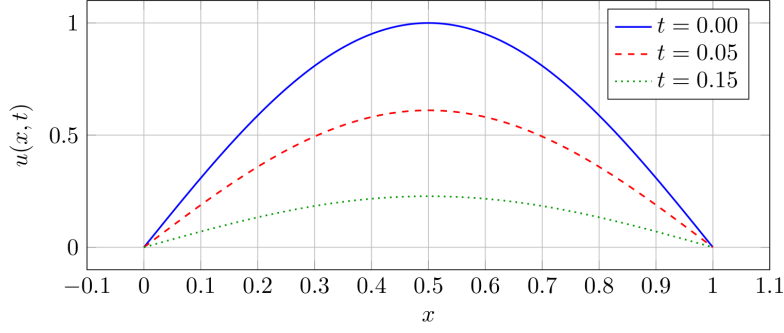


Figure 1: Temporal evolution of the temperature profile $u(x, t)$ for $\alpha = 1$. The amplitude decays exponentially at a rate of $\pi^2\alpha$.

Table 1: Maximum error and experimental order of convergence p as a function of N .

N	Δx	E_N	$p = \log_2(E_{N/2}/E_N)$
20	5.00×10^{-2}	8.21×10^{-3}	—
40	2.50×10^{-2}	2.05×10^{-3}	2.00
80	1.25×10^{-2}	5.13×10^{-4}	2.00
160	6.25×10^{-3}	1.28×10^{-4}	2.00

5 Conclusion

We have derived, implemented, and numerically validated the explicit FTCS method for solving the heat equation. The von Neumann stability condition $r \leq 1/2$ proved to be necessary and sufficient for stable solutions, and the empirical order of convergence aligns with the theoretical expectation of $\mathcal{O}(\Delta x^2)$. However, for practical applications requiring fine spatial resolution, an implicit method—such as the Crank-Nicolson scheme—is preferable to the explicit approach presented here due to the condition $\Delta t \propto \Delta x^2$, as it is unconditionally stable and allows for larger time steps.